Marc Alexis

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Education

University of Central Florida Orlando, FL
B.S. in Computer Science; Minor in Robotics *May 2028*
Coursework: Data Structures & Algorithms, Operating Systems, Computer Organization, Discrete Structures, Calculus II

Organizations: Knight Hacks, NSBE, ColorStack

Projects

Guiding Hands | *Python, OpenCV, MediaPipe, JavaScript, WebSockets, React, Jetson Orin Nano*
- Built a real-time computer vision pipeline using **Python**, **OpenCV**, and **MediaPipe** to map **4** hand gestures to directional controls for a rehabilitation game.
- Bridged **Python**-based gesture recognition to a **3-level** browser game through a **UDP-to-WebSocket** relay, enabling real-time movement updates at up to **30 Hz**.
- Implemented **3-pose** calibration, gesture validation, and tracking-loss safeguards that halt movement when frames exceed **350 ms**.
- Deployed the full pipeline on an **NVIDIA Jetson Orin Nano**, running vision inference and game rendering on-device.

SpeakLab AI | *React Native, TypeScript, Firebase, OpenAI API, Railway*
- Built an AI public speaking coach in **React Native** that records speech, transcribes audio, and generates personalized feedback.
- Validated analysis across **150+** recorded speeches and **10+** metrics, including pacing, pauses, filler words, and vocabulary.
- Generated feedback in **under 2.5 seconds** via an **OpenAI API** pipeline integrated with **Firebase** and **Railway**.

Road Report | *TypeScript, React, Next.js, TigerData, ElevenLabs, REST APIs*
- Built a weighted phrase-scoring classifier for **5** hazard categories, achieving **92%** accuracy across **100+** reports, including noisy voice submissions.
- Cut geospatial processing by approx. **67%** via **0.3-mile** incident clustering and geohashing.
- Integrated **TigerData** for geospatial incident storage and **ElevenLabs** for hands-free conversational hazard reporting.
- Implemented a confirm-before-submit **REST API** flow with idempotency protection to prevent duplicate submissions.

ZBXP | *React, JavaScript, Supabase, PostgreSQL, REST APIs*
- Built a gamified productivity platform in **React** for **50+** users with XP progression, streaks, quests, and task tracking.
- Tracked **50+** completed tasks and **5+ hours** of Pomodoro sessions through persistent productivity workflows.
- Modeled users, tasks, guilds, and sessions in **Supabase**/**PostgreSQL**; delivered leaderboards with a **1.8-second** average refresh.

Experience

Security Team Lead Aug 2021 – Aug 2023
Universal Studios Florida *Orlando, FL*
- Led a **12-person** undercover security team supporting events with **30,000–35,000** daily visitors.
- Partnered with the Orlando Police Department to identify and remove **50+** trespass violators.
- Coordinated assignments and real-time incident response, serving as liaison between security staff and park leadership.

Volunteer May 2022 – Jan 2025
Boys & Girls Club *Orlando, FL*
- Provided weekly academic support and homework assistance to **20–30** students.
- Led sports and group activities focused on teamwork, engagement, and social development.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, C, HTML/CSS
Frameworks & Libraries: React, React Native, Next.js, Node.js, Express, Expo, OpenCV, MediaPipe, REST APIs
Databases: PostgreSQL, TigerData, Supabase, Firebase, Firestore
Tools & Platforms: NVIDIA Jetson Orin Nano, WebSockets, Railway, Vercel, Git, GitHub, OpenAI API, ElevenLabs
Software Engineering: Object-Oriented Programming, API Design & Integration